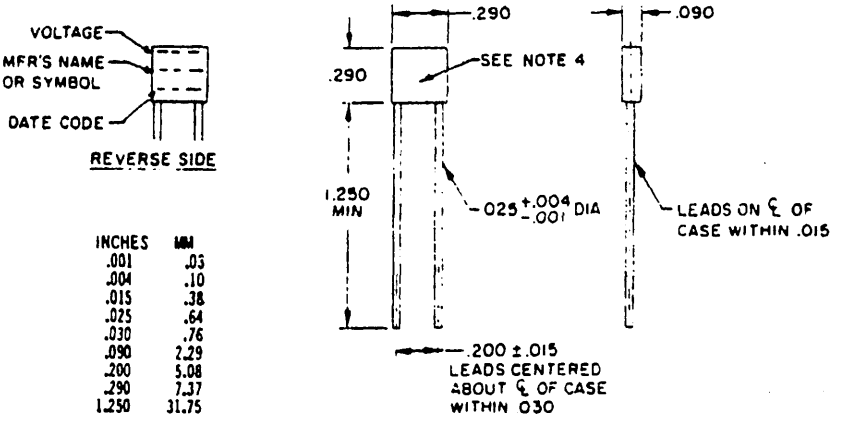
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), STYLE CK06

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-11015.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

INACTIVE FOR NEW DESIGN
after 28 September 1970. Use MIL-C-39014/2



INCHES	MM
.001	.03
.004	.10
.015	.38
.025	.64
.030	.76
.090	2.29
.200	5.08
.290	7.37
1.250	31.75

NOTES:
1. All dimensions in inches.
2. Unless otherwise specified, tolerance is ±.010 (.25 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Type designation marking to be divided into two or three lines; one line marking shall not be used.

FIGURE 1. Style CK06 capacitors.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1.
Case type - Flat plate, molded or pre-formed.
Capacitance value - See table I.
Capacitance tolerance - ±10 percent (K) or ±20 percent (M) as shown in table I.
Rated temperature - -55° to +125° C (although 1,200 pF through 10,000 pF capacitors are identified as B units, they shall have the capability of -55° to +150° C operation, see figure 2.)

Dielectric withstanding voltage:
Dielectric (DWV): Method 301 of MIL-STD-202.
Test voltage - 250 percent of rated voltage.
Body insulation: In accordance with MIL-C-11015.
Test potential - 1,300 volts dc.

Barometric pressure (reduced): Method 105 of MIL-STD-202. 0.82 inch of mercury (80,000 ft).
Test potential - 150 percent of rated voltage.

FSC 5910

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TABLE I. Style CK06 characteristics.

Type designation 1/	Rated voltage	Rated temperature and voltage-temperature limits	Capacitance	Capacitance tolerance
	volts, dc		pF	
CK06BX122K	200	BX	1,200	K
CK06BX152-	200	BX	1,500	K, M
CK06BX182K	200	BX	1,800	K
CK06BX222	200	BX	2,200	K, M
CK06BX272K	200	BX	2,700	K
CK06BX332-	200	BX	3,300	K, M
CK06BX392K	200	BX	3,900	K
CK06BX472-	200	BX	4,700	K, M
CK06BX562K	200	BX	5,600	K
CK06BX682-	200	BX	6,800	K, M
CK06BX822K	200	BX	8,200	K
CK06BX103-	200	BX	10,000	K, M
CK06BX123K	100	BX	12,000	K
CK06BX153-	100	BX	15,000	K, M
CK06BX183K	100	BX	18,000	K
CK06BX223-	100	BX	22,000	K, M
CK06BX273K	100	BX	27,000	K
CK06BX333-	100	BX	33,000	K, M
CK06BX393K	100	BX	39,000	K
CK06BX473-	100	BX	47,000	K, M
CK06BX563K	100	BX	56,000	K
CK06BX683-	100	BX	68,000	K, M
CK06BX823K	100	BX	82,000	K
CK06BX104-	100	BX	100,000	K, M
CK06BX124K	50	BX	120,000	K
CK06BX154-	50	BX	150,000	K, M
CK06BX184K	50	BX	180,000	K
CK06BX224-	50	BX	220,000	K, M
CK06BX274K	50	BX	270,000	K
CK06BX334-	50	BX	330,000	K, M
CK06BX394K	50	BX	390,000	K
CK06BX474-	50	BX	470,000	K, M
CK06BX564K	50	BX	560,000	K
CK06BX684-	50	BX	680,000	K, M
CK06BX824K	50	BX	820,000	K
CK06BX105-	50	BX	1,000,000	K, M

1/ Where applicable, the complete type designation will include an additional symbol to indicate capacitance tolerance.

Insulation resistance (IR): Method 302 of MIL-STD-202, rated voltage applied. 100,000 megohms, min or 1,000 megohm-microfarads, min, whichever is less.

Capacitance (Cap.): Method 305 of MIL-STD-202.
Test frequency: 1 kHz ± 100 Hz.
Test voltage: 1.0 ± 0.2 volt rms.

Dissipation factor (DF): 2.5 percent, max.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100 G's).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 G's).

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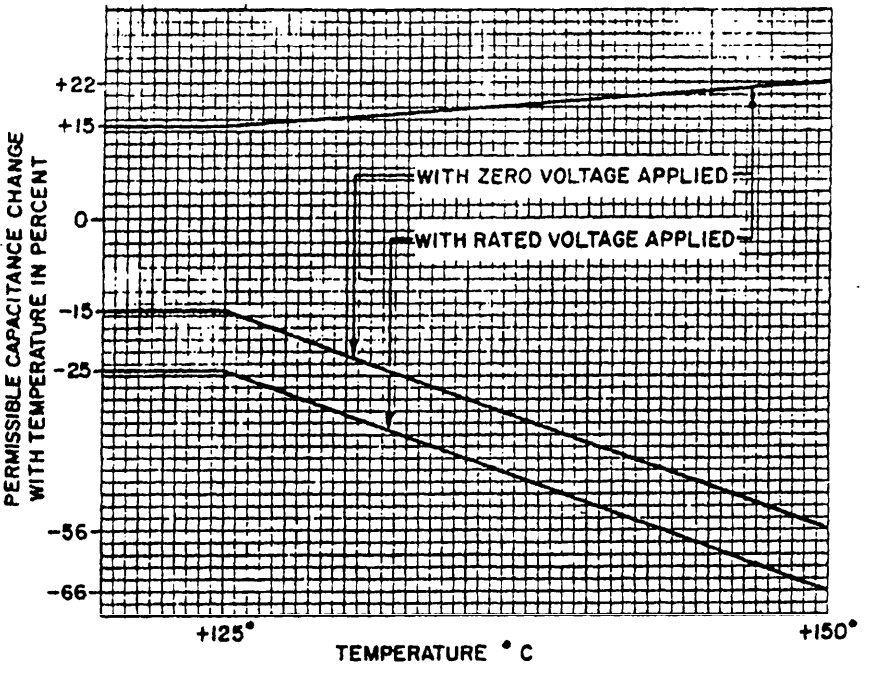


FIGURE 2. Extended voltage-temperature limits (1,200 pF through 10,000 pF capacitors).

Temperature cycling and immersion:
Temperature cycling: Method 102 of MIL-STD-202, condition D, except step 3 shall be at high test temperature.
Immersion: Method 104 of MIL-STD-202, condition B.
IR - Not less than 10 percent of initial requirement.

Salt spray (corrosion): Not applicable.

Terminal strength: In accordance with MIL-C-11015, 5 pounds applied.

Moisture resistance: Method 106 of MIL-STD-202.
DWV - 250 percent of rated voltage.
IR - Not less than 10 percent of initial requirement.
Cap. - Within tolerance of table I value.

Solderability: Method 208 of MIL-STD-202. 2 terminals.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.

Resistance to soldering heat (cont):
IR - 100,000 megohms, min or 1,000 megohm-microfarads, min, whichever is less.
ΔCap. - ±5 percent of initial measurement.
ΔDF - 0.5 percent, max.

Voltage-temperature limits: In accordance with MIL-C-11015 and figure 2 (for 1,200 pF through 10,000 pF capacitors).

Life (at elevated ambient temperature):
Method 108 of MIL-STD-202. 2,000 hours (for qualification); 250 hours (performance check); 1,750 hours (continuation test).
Test potential - 200 percent of rated voltage.
IR - 1,000 megohms minimum or 10 megohm-microfarads, min, whichever is less (at 125° C), and not less than 10 percent of the initial requirement (at 25° C).
DF - 2.5 percent, max (at 25° C).

Marking: In accordance with MIL-C-11015, and as shown on figure 1.

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Supersession data: Items having a "CW" voltage-temperature limit formerly covered by Rev. "B", and earlier issues, of this specification sheet are superseded as indicated in table II.

TABLE II. Cross-reference of substitute items.

Superseded item 1/	Recommended substitute item 1/	Superseded item 1/	Recommended substitute item 1/
CK06CW122K	CK06BX122K	CK06CW392K	CK06BX392K
CK06CW152-	CK06BX152-	CK06CW472-	CK06BX472-
CK06CW182K	CK06BX182K	CK06CW562K	CK06BX562K
CK06CW222-	CK06BX222-	CK06CW682-	CK06BX682-
CK06CW272K	CK06BX272K	CK06CW822K	CK06BX822K
CK06CW332-	CK06BX332-	CK06CW103-	CK06BX103-

1/Where required, the type designation will include comparable capacitance tolerance symbol.

Inactivated styles: For new design it is recommended that units of MIL-C-39014/2 be considered as listed in table III. Although table III lists only the highest failure rate level, lower failure rate levels may be considered. For substitutability of the inactivated styles, capacitors in MIL-C-39014/2 of the same capacitance value, tolerance, rated voltage and temperature coefficient can be used regardless of the failure rate designation.

TABLE III. Cross-reference of inactivated styles.

From inactivated MIL-C-11015	To MIL-C-39014/2 L (2.0) M39014/02-	From inactivated MIL-C-11015	To MIL-C-39014/2 L (2.0) M39014/02-
CK06BX122K	0361	CK06BX333-	0385
CK06BX152K	0362	CK06BX393K	0386
CK06BX152M	0363	CK06BX473-	0387
CK06BX182K	0364	CK06BX563K	0388
CK06BX222K	0366	CK06BX683-	0389
CK06BX222M	0367	CK06BX813K	0390
CK06BX272K	0368	CK06BX104-	0391
CK06BX332K	0369	CK06BX124K	0392
CK06BX332M	0370	CK06BX154-	0393
CK06BX392K	0371	CK06BX184K	0394
CK06BX472K	0372	CK06BX224-	0395
CK06BX472M	0373	CK06BX274K	0396
CK06BX562K	0374	CK06BX334-	0397
CK06BX682K	0375	CK06BX394K	0398
CK06BX682M	0376	CK06BX474-	0399
CK06BX822K	0377	CK06BX564K	0400
CK06BX103K	0378	CK06BX684-	0401
CK06BX103M	0379	CK06BX824K	0402
CK06BX123K	0380	CK06BX105-	0403
CK06BX153-	0381		
CK06BX183K	0382		
CK06BX223-	0383		
CK06BX273K	0384		

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Navy - EC
Air Force - 11

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Navy - AS, OS, SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

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